



Species of Monogenoidea from fish species used in aquaculture in the Peruvian Amazonia

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ABSTRACT

In the present study it is presented a list of species of Monogenoidea with their fish-hosts used in aquaculture in the Peruvian Amazonia. Fish species were obtained from April 2018 to April 2022 from different artificial fish ponds located in “Centro de Investigaciones Fernando Alcantara Bocanegra” (CIFAB) of the “Instituto de Investigaciones de la Amazonía Peruana (IIAP) in Loreto-Peru and from fish ponds of the “Centro de Investigación Carlos Miguel Castañeda Ruiz” from IIAP, San Martín- Peru. A total list of forty-eight species is presenting in the present study: one species from *Arapaima gigas* (Schinz); four species from *Colossoma macropomum* (Cuvier); five species from *Myloplus schomburgkii* (Jardine); two species from *Piaractus brachipomus* (Cuvier); 19 species from *Pygocentrus nattereri* (Kner); one species from *Prochilodus nigricans* (Spix & Agassiz); five species from *Brycon amazonicus* (Spix & Agassiz); one species from *Chaetobranchius flavescens*; three species from *Astronotus ocellatus* (Agassiz); two species from *Cichla monoculus* (Agassiz); four species from *Pterigoplichthys pardalis* (Castelnau); three species from *Calophysus macropterus* (Lichtenstein); and three species from *Pseudoplatystoma punctifer* (Castelnau).

1. Introduction

The knowledge of the biodiversity of Monogenoidea parasites in fish-hosts from South America have been explored from different authors and registered in catalogues and checklists: Nasir and Fuentes (1983) for Latin America; Kohn and Santos (1989); Thatcher (1991); Boeger (2006) for the Neotropical Region; Kohn and Santos (1989); Kohn and Paiva (2000) and Cohen et al. (2013) for South American countries; Luque et al. (2016) for Peru. (See Figs. 1–13).

Fish species commercialized in the markets of Loreto, Peru come from fishing in natural environments and others come from aquaculture production. This activity has been experiencing a constant growth per year. Within the main species cultivated in the peruvian Amazonia and offered in the markets of the cities stands out species of the Arapaimidae, Serrasalminidae, Bryconidae, Cichlidae, Loricaridae and Pimelodidae (García Dávila et al., 2018).

The aim of this paper is to update the list of species and hosts of South American Monogenoidea presenting for the first time a specific checklist of this group of parasites from fish-hosts used in aquaculture in the

Peruvian Amazonia.

2. Material and methods

Fish host species were obtained from April 2018 to April 2022 from different artificial fish ponds located in “Centro de Investigaciones Fernando Alcantara Bocanegra” (CIFAB) of the “Instituto de Investigaciones de la Amazonía Peruana (IIAP) in Loreto-Peru and from fish ponds of the “Centro de Investigación Carlos Miguel Castañeda Ruiz” from IIAP, San Martín-Peru.

Species collected in CIFAB-IIAP, Loreto Peru were: the arapaima *Arapaima gigas*, the black-band myleus *Myloplus schomburgkii*, the black pacu *Colossoma macropomum*, the red-bellied pacu *Piaractus brachipomus*, the red-bellied piranha *Pygocentrus nattereri*, the black prochilodus *Prochilodus nigricans*, the dorado *Brycon amazonicus*, the giant basket-mouth cichlid *Chaetobranchius flavescens*, the oscar *Astronotus ocellatus*, the peacock bass *Cichla monoculus* and the spotted tiger shovelnose catfish *Pseudoplatystoma punctifer*. Species collected from “Centro de Investigación Carlos Miguel Castañeda Ruiz”, San Martín-Peru were: the

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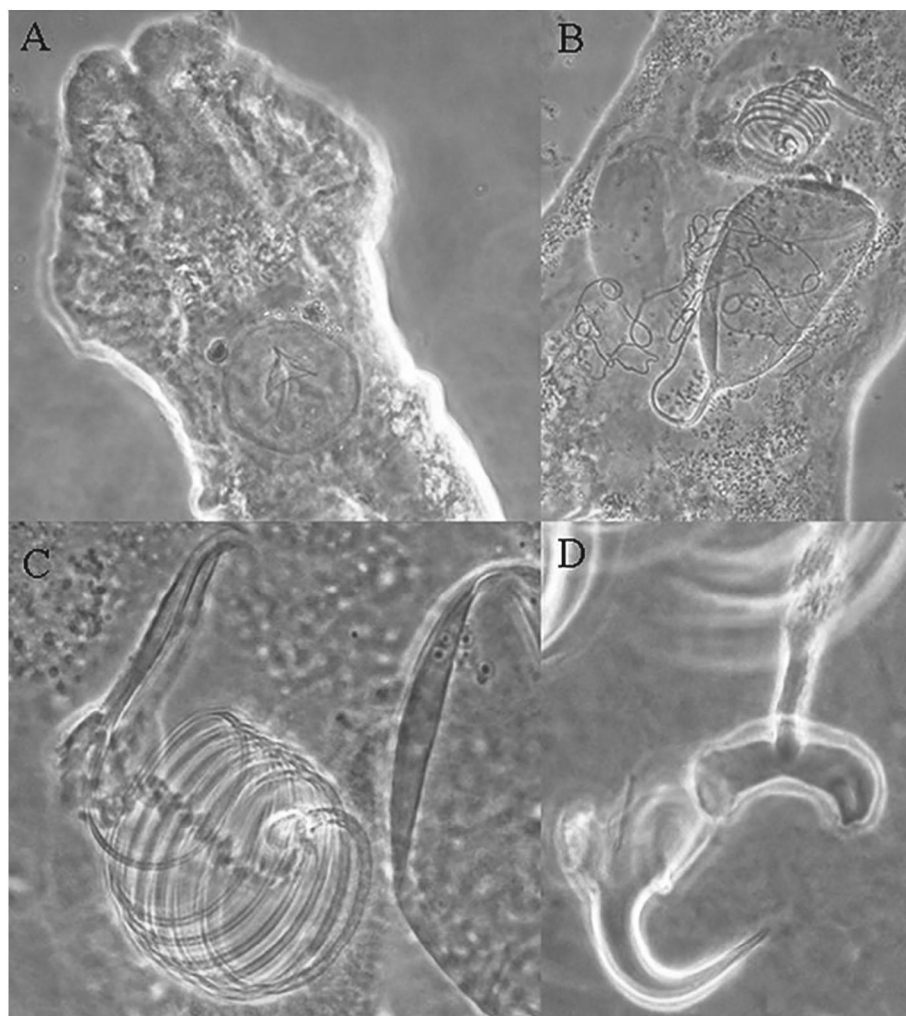


Fig. 1. *Dawestrema cycloancistrum*. A. anterior part of body showing the head. B. Copulatory complex and egg. C. Copulatory complex: Male copulatory organ (MCO) and accessory piece. D. Ventral bar and ventral anchor.

Amazon sailfin catfish *Pterygoplichthys pardalis*, and the vulture catfish *Calophysus macropterus*. Fish host were identified on the basis of morphological characters according to García Dávila et al. (2018).

Fish were kept alive until their parasitological examination. In the laboratory, gill archers were removed and placed in vials containing hot water (65 °C). Each vial was shaken vigorously and 96% ethanol was added. The content of each vial was examined using a dissecting microscope and helminths were removed from the gills or sediment using Hoyer's medium to determine the haptor structures (bars, anchors and hooks) and male copulatory organ (MCO) and the accessory piece. Others specimens were stained with Gomori's trichrome (Humason 1979; Boeger and Vianna 2006) and mounted in Canada balsam to determine internal organs. Some specimens were preserved in ethanol 96% for future DNA analyses. All procedures were conducted in the "Laboratorio de Parasitología y Sanidad Acuicola" of the "Instituto de Investigaciones de la Amazonía Peruana" (IIAP) in Iquitos, Peru. Pictures of the parasites were taken by using a digital camera LEICA ICC50 W attached to a microscope LEICA DM750. Data were processed by using a LEICA Application Suite (LAS) EZ.

3. Results

3.1. List of Monogenoidea by host

OSTEOGLOSSIFORMES

ARAPAIMIDAE

Host: *Arapaima gigas* (Schinz, 1822)

Dawestrema cycloancistrum Price & Nowlin, 1967

Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru

CHARACIFORMES

SERRASALMIDAE

Host: *Colossoma macropomum* (Cuvier, 1816)

Anacanthorus spathulatus Kritsky, Thatcher & Kayton, 1979

Notozothecium janauachensis Belmont-Jegu, Domingues, & Martins, 2004

Mymarothecium boegeri Cohen & Kohn, 2005

Mymarothecium iapensis Morey, Aliano & Grández, 2019

Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru

Host: *Myloplus schomburgkii* (Jardine, 1841)

Anacanthorus camposbae Morey, Aliano & Grández, 2019

Anacanthorus carmenrosae Morey, Aliano & Grández, 2019

Anacanthorus pedanophallus Kritsky, Boeger & Van Every, 1992

Notozothecium bethae Kritsky, Boeger & Jegu, 1996

Notozothecium nanayensis Morey, Aliano & Grández, 2019

Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru

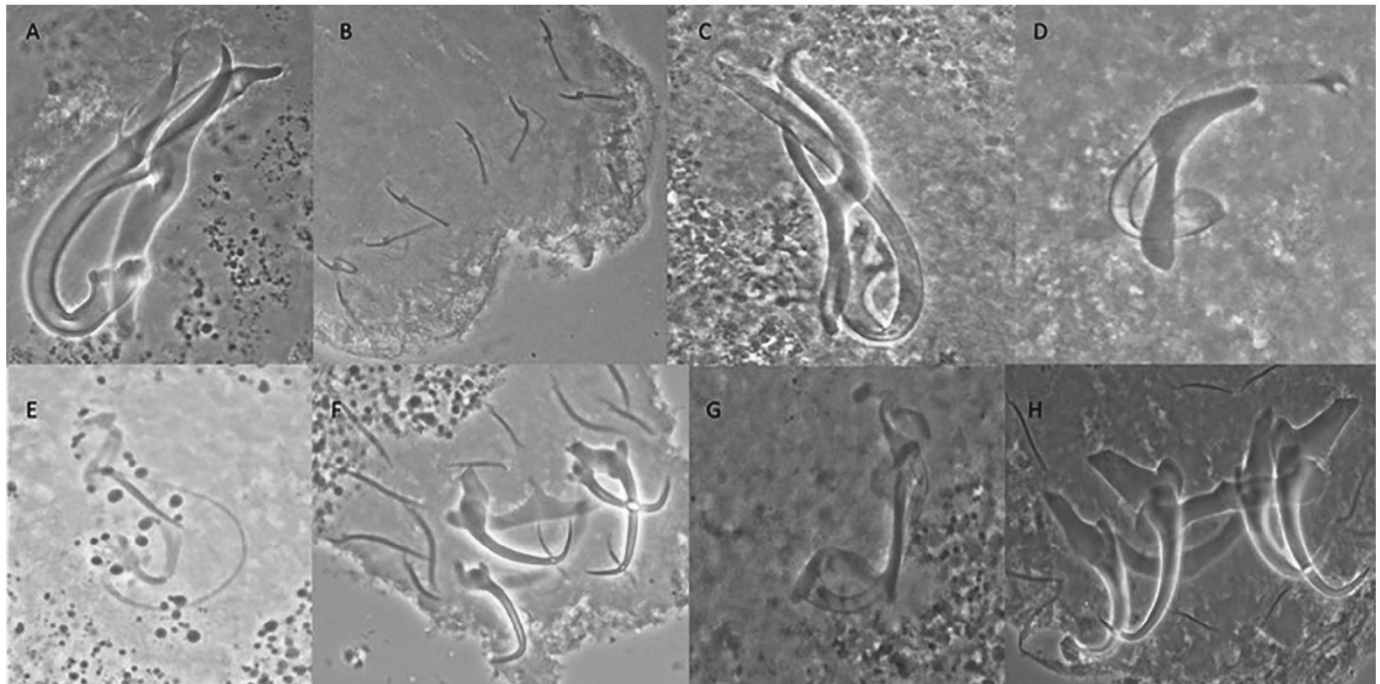


Fig. 2. A. Copulatory complex of *Anacanthorus camposbae*. B. Haptor showing hooks. C. Copulatory complex of *Anacanthorus carmenrosae*. D. Copulatory complex of *Anacanthorus pedanophallus*. E. Copulatory complex of *Notozothecium bethae*. F. Haptor. G. Copulatory complex of *Notozothecium nanayensis*. H. Haptor.

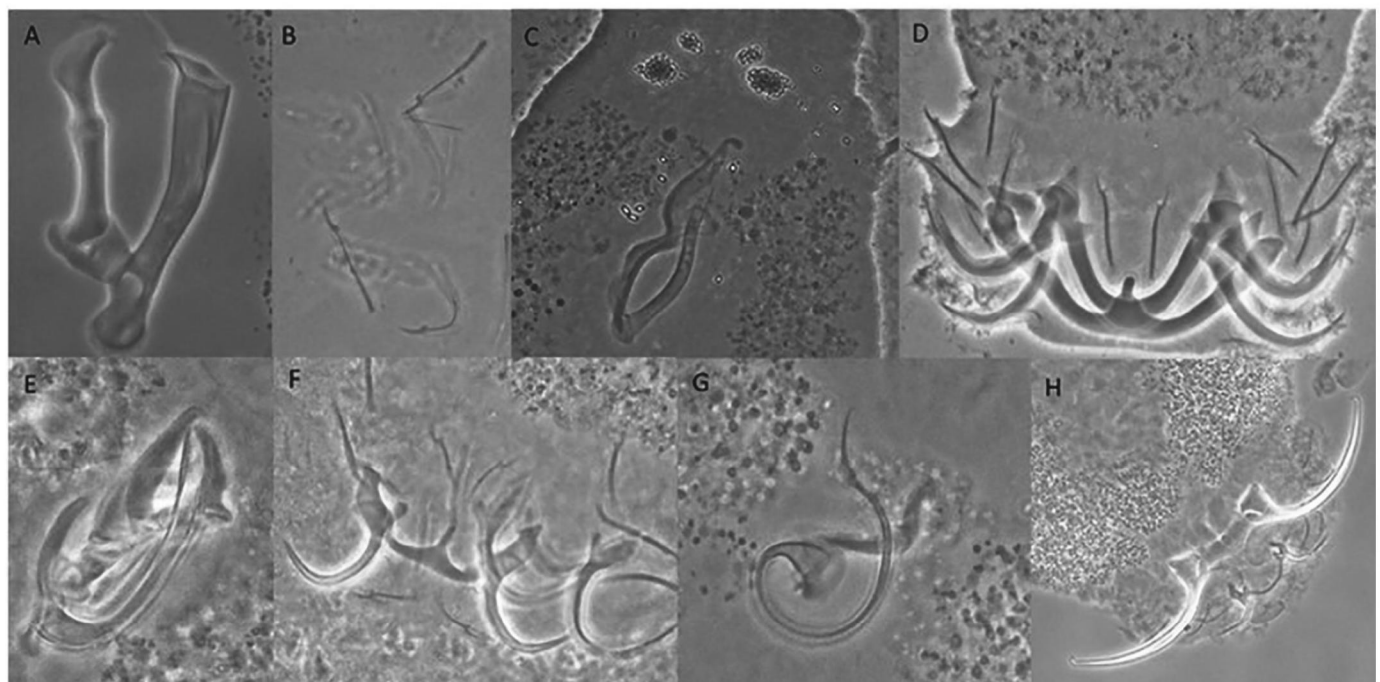


Fig. 3. A. Copulatory complex of *Anacanthorus spatulatus*. B. Hooks. C. Copulatory complex of *Mymarothecium boegeri*. D. Haptor. E. Copulatory complex of *Mymarothecium iipensis*. F. Haptor. G. Copulatory complex of *Notozothecium janauchensis*. H. Haptor.

Host: *Piaractus brachipomus* (Cuvier, 1818)

Anacanthorus penilabiatus Boeger, Husak & Martins, 1995

Mymarothecium viatorum Boeger, Piasecki & Sobecka, 2002

Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru

Host: *Pygocentrus natterei* Kner, 1858

Amphithecium camelum Boeger and Kritsky, 1988

Amphithecium cataloensis Boeger and Kritsky, 1988

Amphithecium falcatum Boeger & Kritsky, 1988

Amphithecium junki Boeger & Kritsky, 1988

Amphithecium calycinum Boeger & Kritsky, 1988

Anacanthorus amazonicus Van Every & Kritsky, 1992

Anacanthorus penilabiatus Boeger, Husak & Martins, 1995

Anacanthorus ramosissimus Van Every & Kritsky, 1992

Anacanthorus reginae Boeger & Kritsky, 1988

Anacanthorus scapanus Van Every & Kritsky, 1992

Anacanthorus stachophallus Kritsky, Boeger & Van Every, 1992

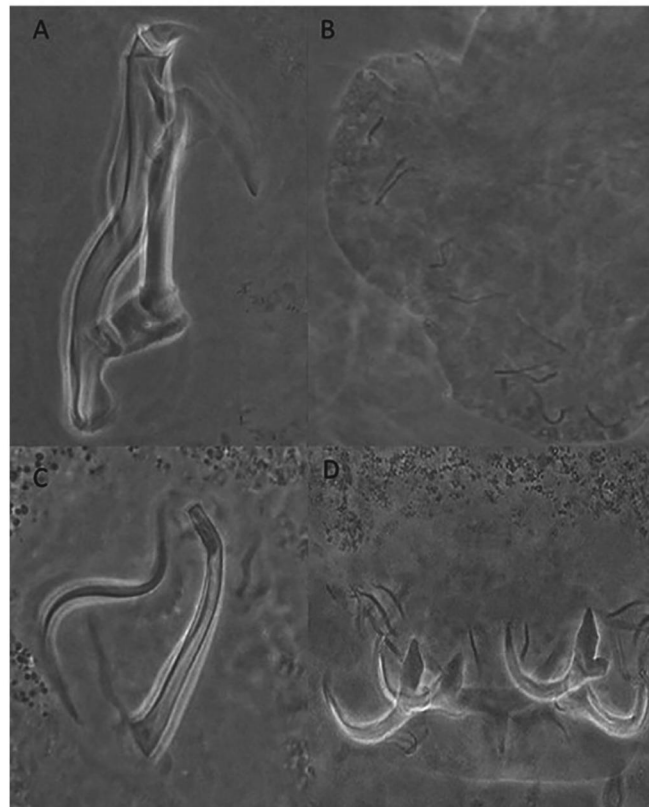


Fig. 4. Copulatory complex of *Anacanthorus penilabiatu*. B. Haptor. C. Copulatory complex of *Mymarothecium viatorum*. D. Haptor.

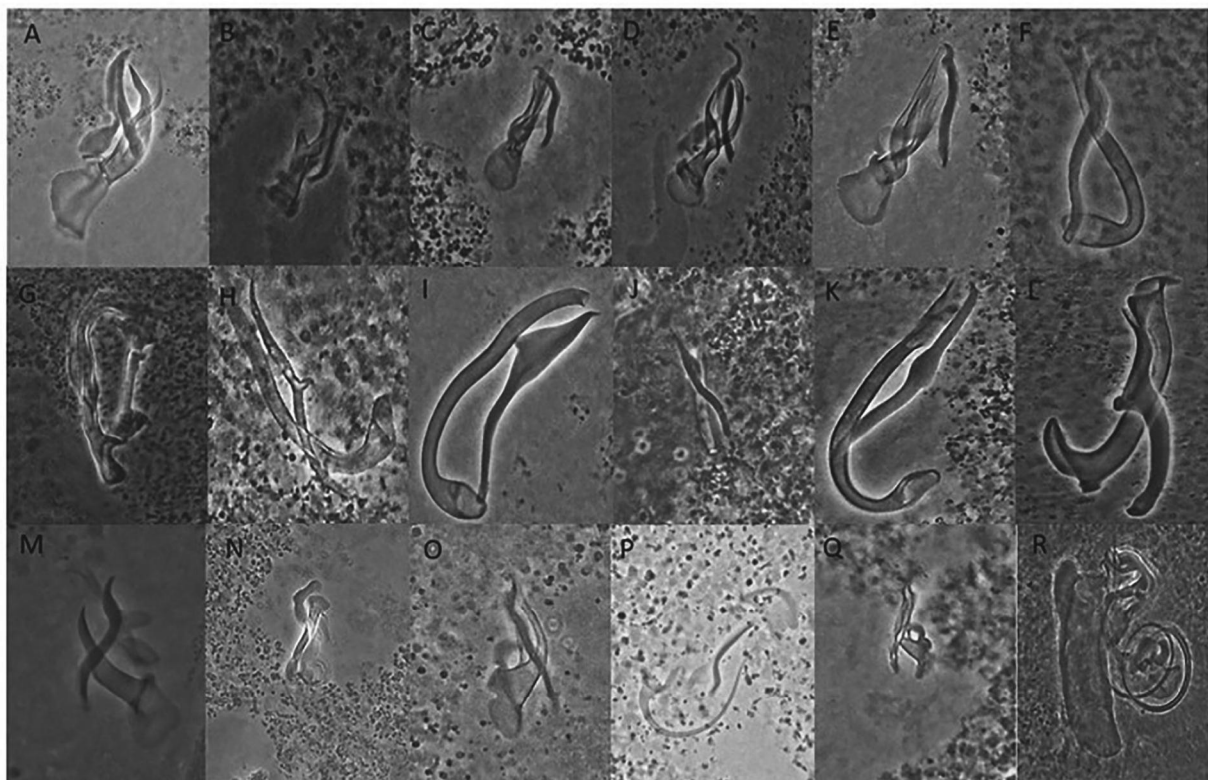


Fig. 5. Copulatory complex of species of Monogeneidae reported in *Pygocentrus nattereri*. A. *Amphithecium camelum*. B. *A. cataloensis*. C. *Amphithecium falcatum*. D. *Amphithecium junki*. E. *Amphithecium calycinum*. F. *Anacanthorus amazonicus*. G. *Anacanthorus penilabiatu*. H. *Anacanthorus ramosissimus*. I. *Anacanthorus reginae*. J. *Anacanthorus scapanus*. K. *Anacanthorus stochophallus*. L. *Anacanthorus thatcheri*. M. *Enallothecium aegidatum*. N. *Mymarothecium viatorum*. O. *Nototheciodes llewellyni*. P. *Notothecium minor*. Q. *Notothecium* sp. R. *Rhinoxenus piranthus*.

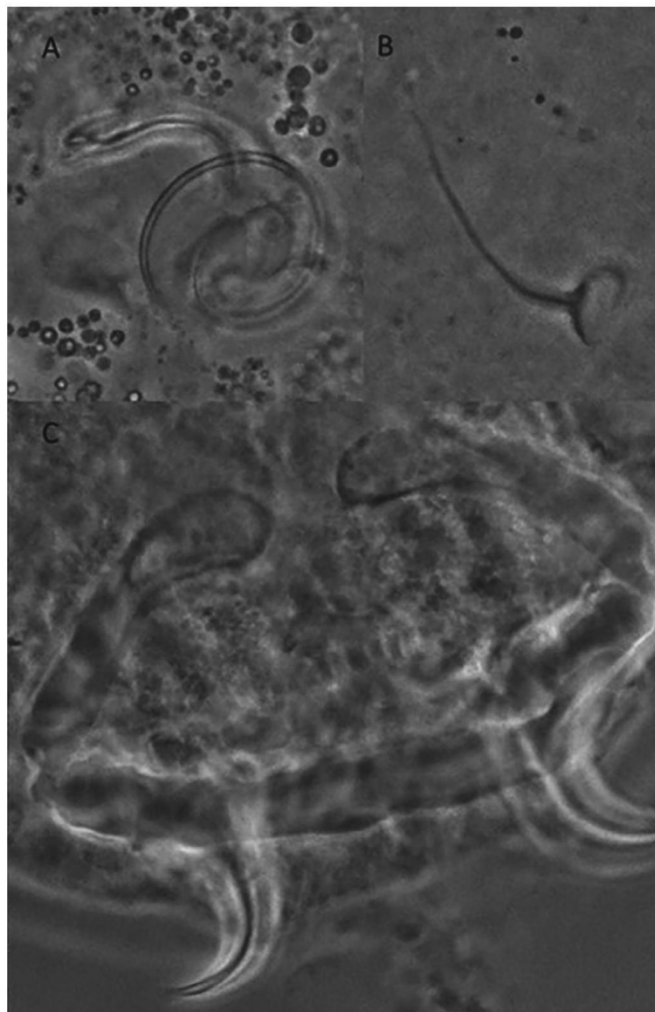


Fig. 6. A. Copulatory complex of *Tereancistrum curimba*. B. Vagina. C. Haptor.

Anacanthorus thatcheri Boeger & Kritsky, 1988
Enallothecium aegidatum Boeger & Kritsky, 1988
Mymarothecium galeolum Kritsky, Boeger & Jegu, 1996
Mymarothecium viatorum Boeger, Piasecki & Sobecka, 2002
Notothecioides llewellyni Kritsky, Boeger & Jegu, 1997
Notothecium minor Boeger & Kritsky, 1988
Notothecium sp.
Rhinoxenus piranhus Kritsky, Boeger & Thatcher, 1988
Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru
PROCHILODONTIDAE
Host: *Prochilodus nigricans* Spix & Agassiz, 1829
Tereancistrum curimba Lizama, Takemoto & Pavanelli, 2004
Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru
Host: *Brycon amazonicus* (Spix & Agassiz, 1829)
Anacanthorus femoris Morey, Sol & Cachique, 2021
Anacanthorus kukamensis Morey, Sol & Cachique, 2021
Anacanthorus rarus Morey, Sol & Cachique, 2021
Anacanthorus sabalo Morey, Sol & Cachique, 2021
Anacanthorus spiralcirrus Kritsky, Thatcher & Kayton, 1979.
Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru
CICHLIFORMES
CICHLIDAE
Host: *Chaetobranchius flavescens* Heckel, 1840
Sciadicleithrum edgari Paschoal, Scholz, Tavares-Dias & Luque, 2016
Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru
Host: *Astronotus ocellatus* (Agassiz, 1831)

Gussevia asota Kritsky, Thatcher & Boeger, 1989
Gussevia astronoti Kritsky, Thatcher & Boeger, 1989
Gussevia rogersi Kritsky, Thatcher & Boeger, 1989
Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru
Host: *Cichla monoculus* Agassiz, 1831
Gussevia undulata Kritsky, Thatcher & Boeger, 1986
Sciadicleithrum ergensi Kritsky, Thatcher & Boeger, 1989
Locality: Fish pond from CIFAB-IIAP, Iquitos-Peru
SILURIFORMES
LORICARIDAE
Host: *Pterigoplichthys pardalis* (Castelnau, 1855)
Heteropriapulus heterotylus (Jogunoori, Kritsky & Venkatanarasaiah, 2004)
Trinigyrus peregrinus Nitta & Nagasawa, 2016
Unilatus unilatus Mizelle & Kritsky, 1967
Unilatus brittani Mizelle, Kritsky, and Crane, 1968
Locality: Fish pond from IIAP, San Martin-Peru
PIMELODIDAE
Host: *Calophysus macropterus* (Lichtenstein, 1819)
Ameloblastella unapi
Demidospermus macropteri
Demidospermus sp.
Locality: Fish pond from IIAP, San Martin-Peru
Host: *Pseudoplatystoma punctifer* (Castelnau, 1855)
Vancleaveus cinnus Kritsky, Thatcher & Boeger, 1986
Vancleaveus fungulus Kritsky, Thatcher & Boeger, 1986
Nanayella sp.
Ameloblastella martinae Mendoza-Palmero, Rossin, Irigoitia & Scholz, 2020
Locality: Fish pond from CIFAB-IIAP, Loreto-Peru

4. Discussion

According to Ortega et al. (2010), from the total number of 1064 valid freshwater fish species recorded for Peru, approximately 800 belong to the Amazonia. Despite this richness, few parasitological studies have focused on the fishes inhabiting their hydrological drainages. Some taxonomical studies and reports on the parasites of fishes from the Peruvian Amazon using morphological and molecular approaches on monogenoids have been conducted: (Luque et al., 2016) reported from freshwater fishes from Peru a total number of 85 monogenoids; Mendoza-Franco et al. (2016) reported six other new species; Acosta et al. (2019) reported five new species; (Mendoza-Palmero et al. (2019) with one species; Morey et al. (2019a) with four new species; Morey et al. (2019b) with one new genus and one new species; Morey et al. (2020) with one species; (Mendoza-Palmero et al. (2020) with one new species; Morey et al. (2021) with five new species; Cayulla-Quispe et al. (2021) with one new species; Morey (2021) with three new species and Feronato et al. (2022) with one new species. With these studies, the number of species of Monogenoidea reported for freshwater fishes from Peru is 114. The list presented in the present study reported 50 species of Monogenoidea and it constitutes the first checklist of monogenoids from fish species used in aquaculture in the Peruvian Amazonia.

For *A. gigas* collected from the Neotropical region, the following monogenoids have been reported: *Dawestrema cycloancistrioides*, *D. cycloancistrum* and *D. punctatum* Cohen et al. (2013). In Peru, high levels of infestation by *D. cycloancistrum* affecting specimens of *A. gigas* from fish-farming have been reported by Luque and Iannacone (1991); Delgado et al. (2013); Mathews et al. (2013); Serrano-Martínez et al. (2015) and Murrieta Morey et al. (2020). The results of the present study are similar to the authors mentioned above, reporting only to *D. cycloancistrum*.

For *C. macropomum* the monogenoids *Anacanthorus spathulatus*, *Linquadactyloides brinkmanni*, *Mymarothecium boegeri*, *Nozotothecium euzeti*, *Notothecium janauachensis* have been reported for the Neotropical region (Cohen et al., 2013). In Peru, some studies focused on monogenoids

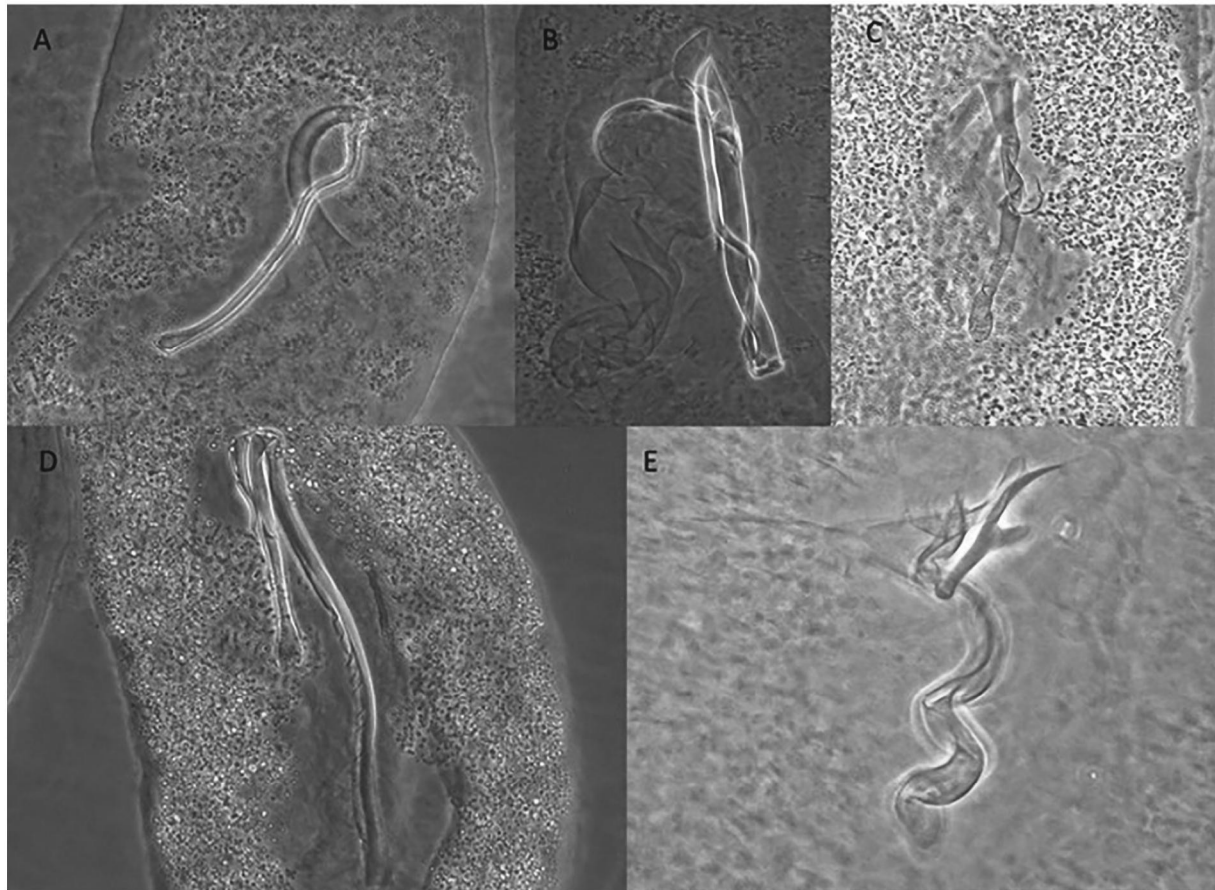


Fig. 7. Copulatory complex of species of Monogenoidea in *Brycon amazonicus*. A. *Anacanthorus femoris*. B. *Anacanthorus kukamensis*. C. *Anacanthorus rarus*. D. *Anacanthorus sabaloi*. E. *Anacanthorus spiralocirrus*.

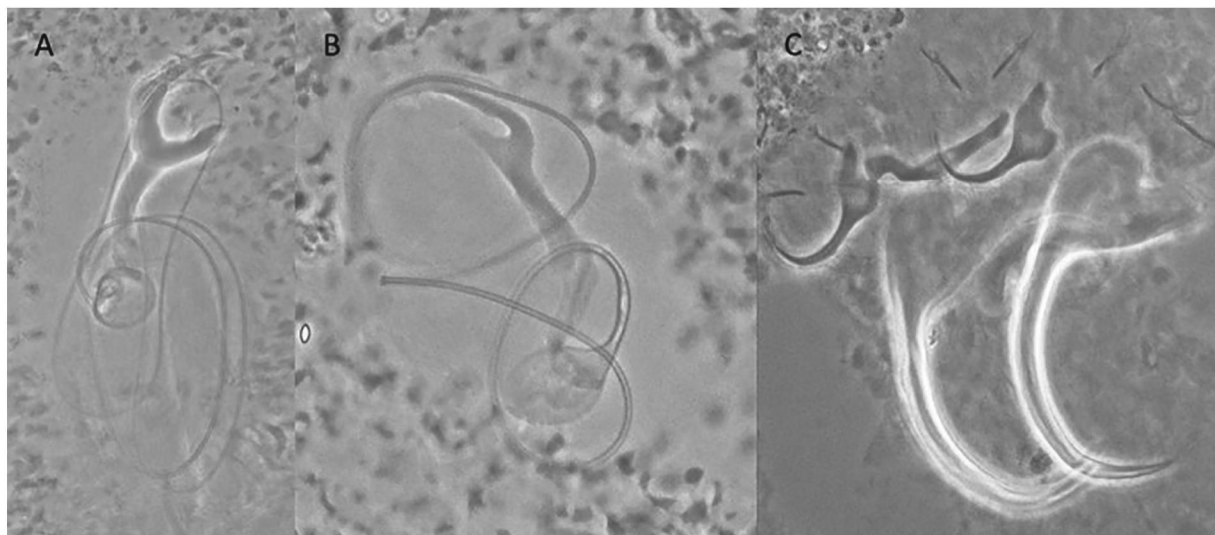


Fig. 8. *Sciadicleithrum edgari* from *Chaetobranchius flavescens*. A, B. Copulatory complex. C. Haptor.

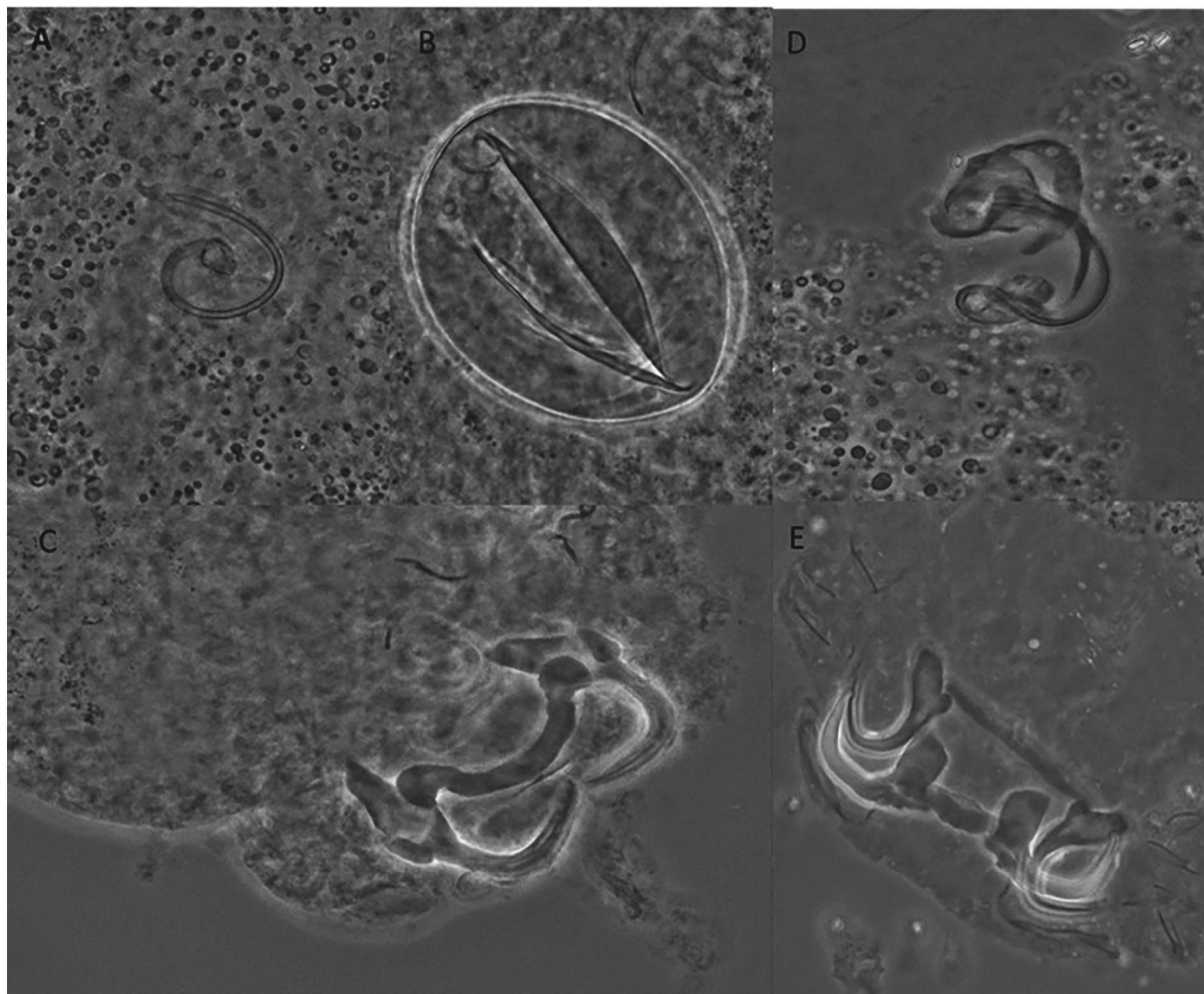


Fig. 9. *Sciadicleithrum ergensi* from *Cichla monoculus*. A. Copulatory complex. B. Egg. C. Haptor. D. Copulatory complex of *Gussevia undulata*. E. Haptor.

from *C. macropomum* collected from artificial ponds have been conducted: *A. spathulatus* from a fish pond in Loreto-Peru (Soberon et al., 2014); *M. iiapensis* Morey et al. (2021); *M. tantaliani* (Cayulla-Quispe et al., 2021). In the present study we report the species *A. spathulatus*, *M. boegeri*, *M. iiapensis* and *N. janauachensis*.

The first record of a monogenoid parasitizing *M. schomburgkii* was cited by Gonzales et al. (2016) identifying *N. bethae* in the gills of specimens from an artificial pond of Loreto-Peru. The second register was made by Morey et al. (2019b) who discovered and described three new species: *A. camposbaeae*, *A. carmenrosae* and *N. nanayensis*. In the present study, the same species reported by Morey et al. (2019a) were found.

For *P. brachypomus* two species have been cited from Brazil: *A. penilabiatus* and *M. viatorum* (Cohen et al., 2013). In Peru *A. spatulatus*, *A. penilabiatus* and *M. viatorum* have been cited from fishes collected in a fish pond from the Peruvian Amazon (Gonzales et al., 2022). In the present study, the same species mentioned above with exception of *A. spatulatus* were found on individuals of *P. brachypomus*.

For *P. nattereri* there were cited the following species: *A. brachycirrum*, *A. calycinum*, *A. camelum*, *A. catalaoensis*, *A. falcatum*, *A. junki*, *A. microphallum*, *A. anacanthorus*, *A. brazilensis*, *A. maltai*, *A. neotropocalis*, *A. reginae*, *A. rondonensis*, *A. stachophallus*, *A. thatcheri*, *Calpidothecioides orthus*, *Calpidothecium crescentis*, *Calpidothecium serrasalmus*, *E. aegidatum*, *M. galeolum*, *N. mizellei*, *N. minor*, *N. penetrarum*, *Pithanothecium amazonensis*, *Pithanothecium piranhush* and *R. piranhush* (Cohen et al., 2013). In the present study, we reported for the first time

in Peru 19 species of Monogenoidea from *P. nattereri* cultivated in an artificial fish pond.

For *P. nigricans* from Peru, the following monogenoids were reported: *Rhinonastes pseudocapsaloideum* from the gills and nasal cavity of *P. nigricans* collected in a fish farm in Loreto, Peru (Delgado et al., 2013), Piña et al. (2017) reported to *Apedunculata discoidea*, *T. curimba* and *T. toksonum* from *P. nigricans* collected in natural environment and a fish pond. In the present study the only monogenoid reported was *T. curimba*.

For *B. amazonicus* the monogenoids *A. spiralcirrus*, *Jainus amazonensis*, *Tereancistrum kerri* and *Trinibaculum brazilensis* were reported for fishes from Brazil (Cohen et al., 2013). Morey et al. (2021) described five new species of Monogenoidea from the gills of *B. amazonicus* collected in natural environment. In the present study, all species recorded by Morey et al. (2021) were found from specimens of *B. amazonicus* collected in a fish pond in the Peruvian Amazonia.

The first study about the parasitic fauna of *C. flavescens* was conducted by Tavares-Dias et al. (2017), who found the monogenoids *G. spilocirra*, and *G. elephus*. In Brazil, specimens of *C. semifasciatus* from Peru were parasitized by *G. tucunarensis* (Mathews et al., 2013). In the present study, we reported the presence of *Sciadicleithrum edgari*, a parasite described for the first time from the gills of *Satanoperca jurupari* collected in the Araguari River, State of Amapá, northern Brazil (Tavares-Dias et al., 2017). In the present study, *S. edgari* is cited in a new host and this is the first report in the Peruvian Amazonia of a monogenoid collected from the gills of *C. flavescens*.

For *A. ocellatus* the monogenoids *G. asota*, *G. astronoti*, *G. rogersi* have been cited parasitizing fishes from Brazil (Cohen et al., 2013). For Peru,

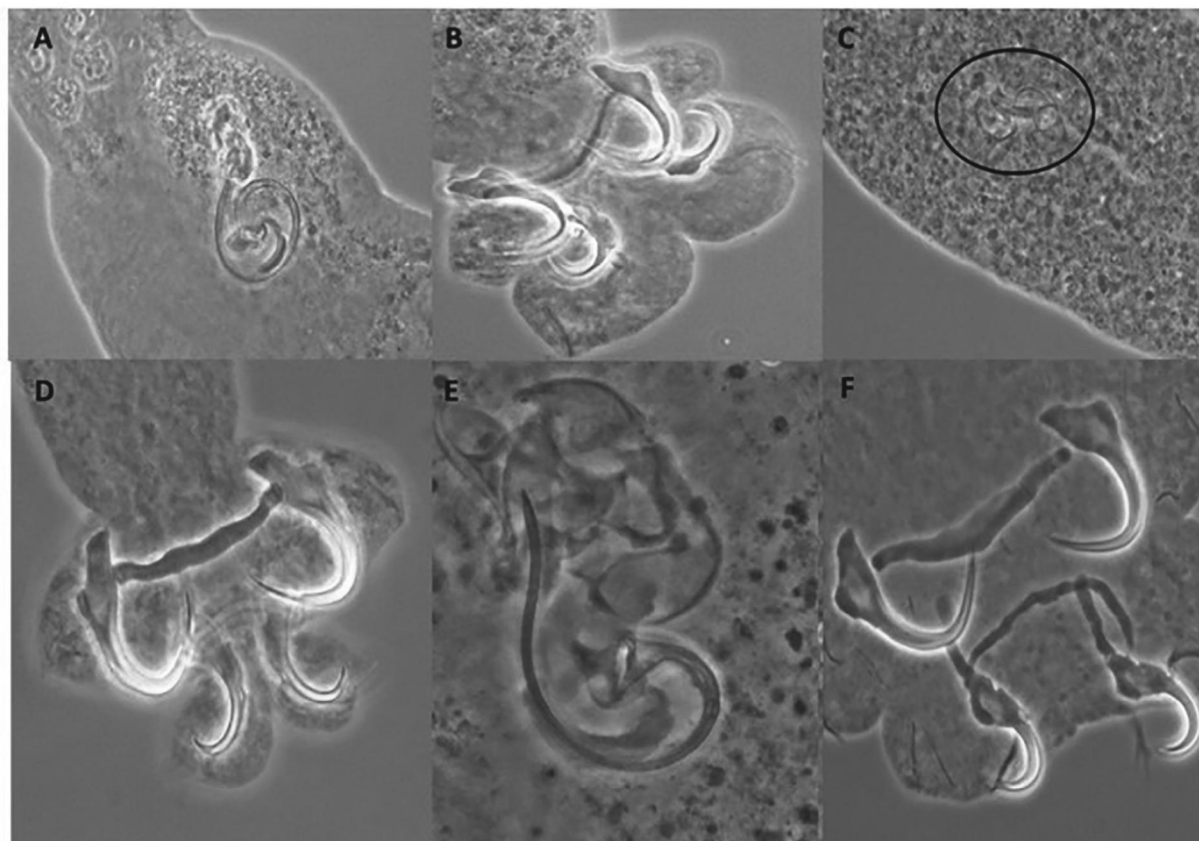


Fig. 10. Species of Monogeneoidea from *Astronotus ocellatus*. A. Copulatory complex of *Gussevia rogersi*. B. Haptor. C. Copulatory complex of *Gussevia asota*. D. Haptor. E. Copulatory complex of *Gussevia astronoti*. F. Haptor.

the only record is for *G. asota* parasitizing specimens from a fish farm in the Peruvian Amazonia with high rates of mortality (Mathews et al., 2013). In the present study, the three species cited by Cohen et al. (2013) were found from fishes cultivated in a fish farm, being *G. astronoti* and *G. rogersi* cited for the first-time parasitizing specimens from *A. ocellatus* from Peru.

For *C. monoculus*, three monogenoids have been cited in Peru: *G. longihaptor*, *G. undulata* and *Tucunarella cichlae*. For Brazil, records about monogenoids parasitizing *C. monoculus* reported the presence of *Gussevia* sp., *G. arilla*, *G. longihaptor*, *G. tucunarensis*, *G. undulata*, *Sciadicleithrum* sp., *S. ergensi*, *S. umbilicum*, *S. uncinatum* and *Tucunarella cichlae* (Santana et al., 2017) and *G. arilla*, *G. longihaptor*, *G. tucunarensis*, *S. ergensi*, *S. umbilicum* and *S. uncinatum* (Oliveira et al., 2019). In the present study, only two species of monogenoids have been cited: *G. undulata* and *S. ergensi* from the gills of *C. monoculus*.

For *P. pardalis*, three species have been cited: *Unilatus unilatus*, *Heteropriapulus heterotylus* and *Trinigyrus mourei* from specimens collected in the Brazilian Amazonia (Porto, 2017). In the present study there were identified the monogenoids *U. unilatus*, *U. brittani*, *H. heterotylus* and *T. peregrinus*.

For *C. macropterus* the following monogenoids have been reported: *Ameloblastella unapi* and *Demidospermus macropteri* from specimens collected in Peru; and *Pavanelliella pavanelli* from specimens collected in Brazil (Cohen et al., 2013). In the present study, *A. unapi*, *Demidospermus* sp. and *D. macropteri* were found in *C. macropterus* collected in the Peruvian Amazonia.

To date, for *P. punctifer*, there are no reports of monogenoids. Reports on the species of Monogeneoidea from *Pseudoplatystoma* spp. are cited by Cohen et al. (2013) who found *Amphocleithrum paraguayensis*,

Pavanelliella pavanelli and *Vancleaveus fungulus* from the gills of *P. corruscans*; *Phanerotecioideus agostinhoi*, *V. fungulus*, *Scleroductus angularis* from *P. fasciatus*; *V. fungulus* from *P. tigrinum* and *Amphocleithrum paraguayensis* from *Pseudoplatystoma* sp. In the present study, from specimens of *P. punctifer* collected in the Peruvian Amazonia, there were found *V. cicinnus*, *V. fungulus*, *Ameloblastella martinae* and *Nanayella* sp.

CRediT authorship contribution statement

Germán Augusto Murrieta Morey: Conceptualization, Methodology, Validation, Investigation, Writing – original draft, Supervision. **Carlos Alfredo Tuesta Rojas:** Methodology, Investigation, Writing – review & editing. **Luciano Alfredo Rodríguez Chu:** Investigation, Formal analysis. **Harvey Satalaya Arellano:** Investigation, Formal analysis. **Gabriela Sandy Culcos Figeroa:** Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

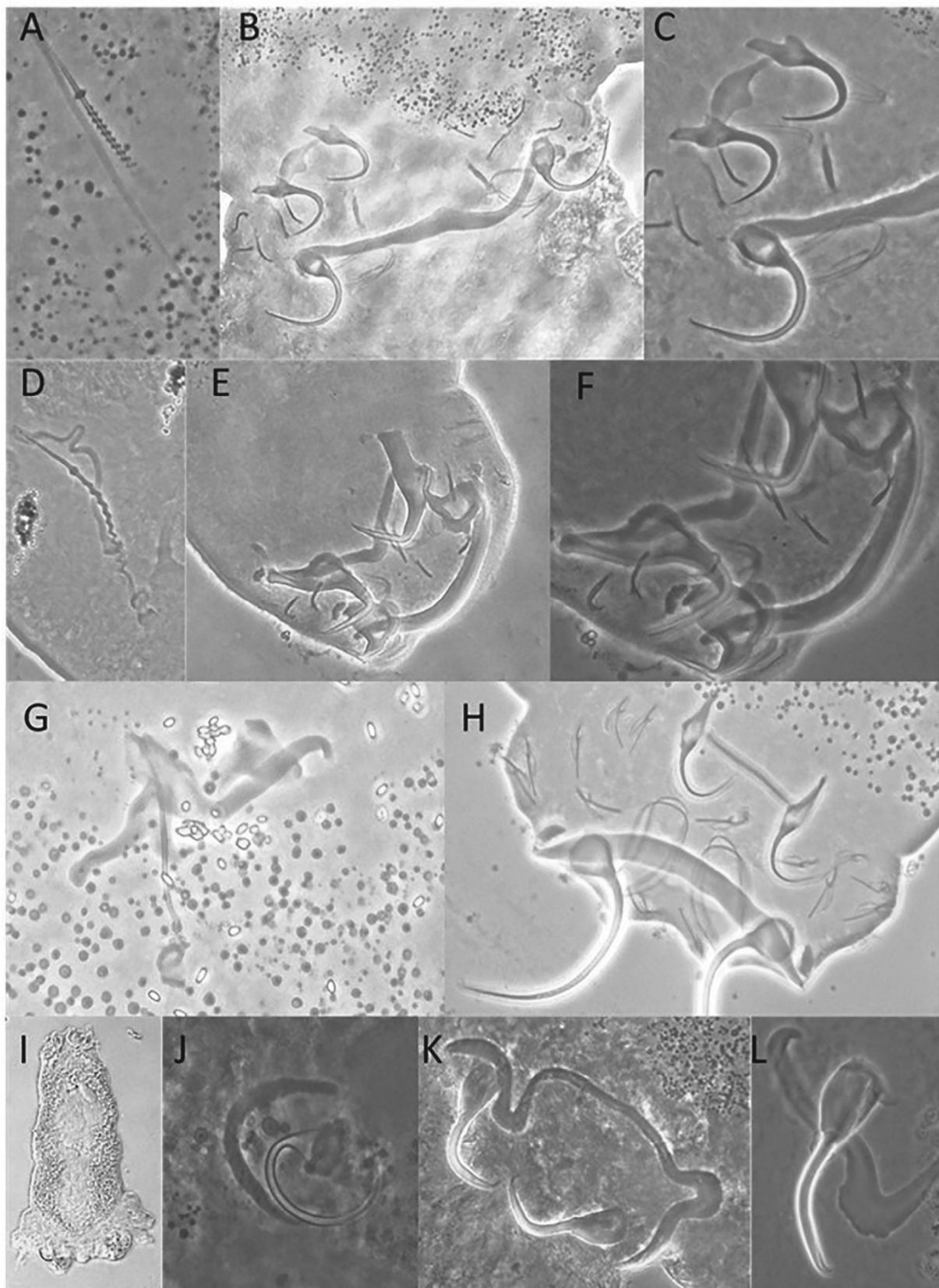


Fig. 11. Species of Monogeneoida from *Pterigoplichthys pardalis*. A. Copulatory complex of *Unilatus brittani*. B, C. Haptor. D. Copulatory complex of *Unilatus unilatus*. E, F. Haptor. G. Copulatory complex of *Heteropriapulus heterotylus*. H. Haptor. I. Body of *Trinigyryus peregrinus*. J. Copulatory complex. K. Haptor. L. Anchor.

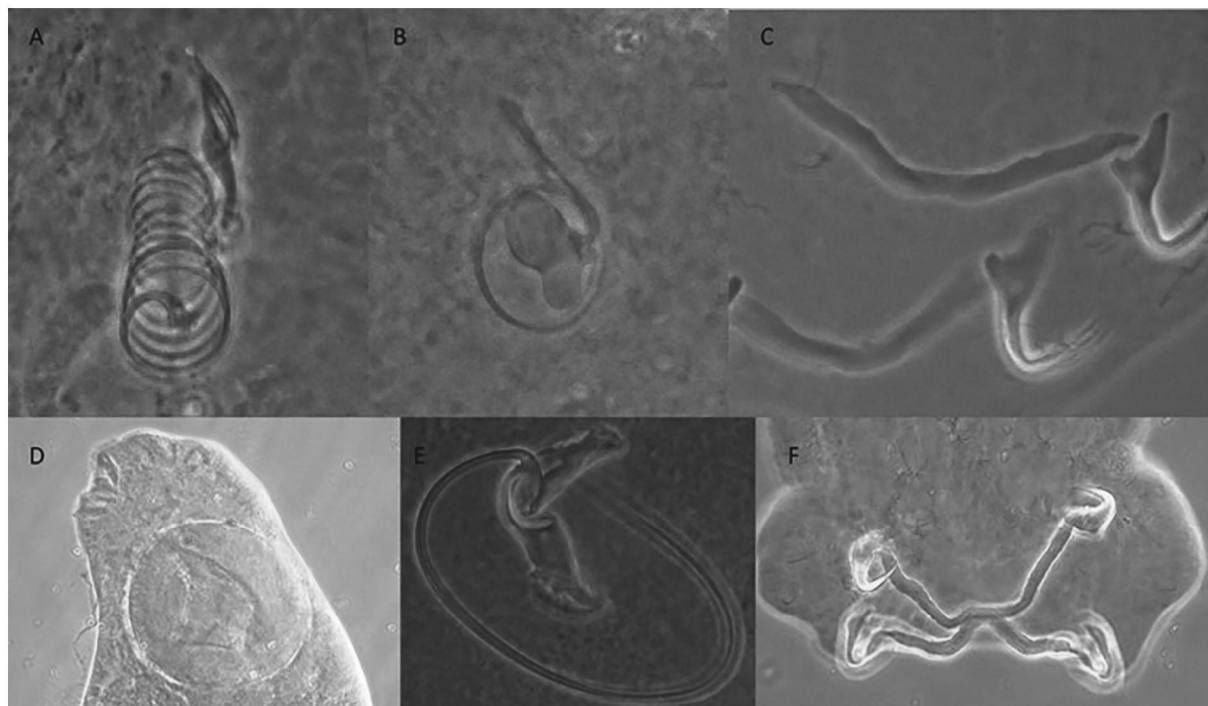


Fig. 12. Species of Monogeneoidea from *Calophrys macropterus*. A. Copulatory complex of *Ameloblastella unapi*. B. Copulatory complex of *Demidospermus macropteri*. C. Haptor. D. Anterior part of *Demidospermus* sp. E. Copulatory complex. F. Haptor.

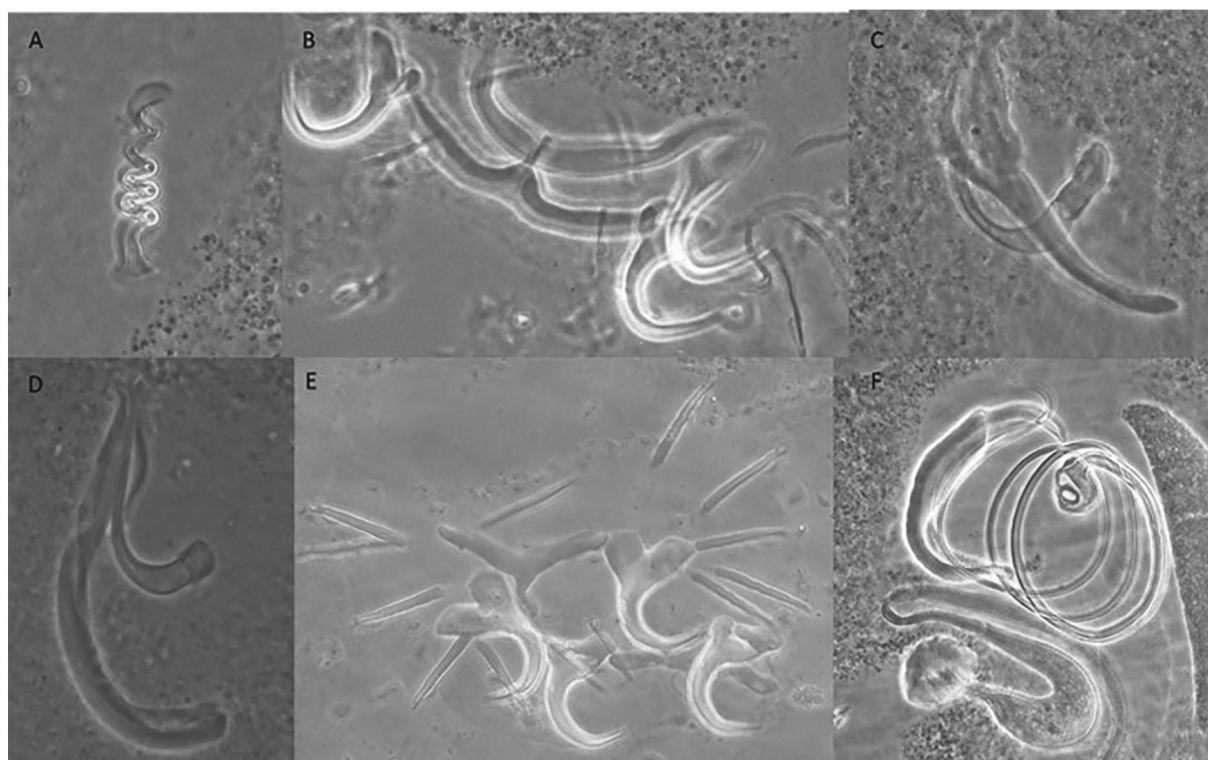


Fig. 13. Species of Monogeneoidea from *Brachyplatystoma punctifer*. A. Copulatory complex of *Ameloblastella martinae*. B. Haptor. C. Copulatory complex of *Vancleaveus fungulus*. D. Copulatory complex of *Vancleaveus cincinnus*. E. Haptor. F. Copulatory complex and vagina of *Nanayella* sp.

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